

MAAT v1.7: Gate-Aware Structural Selection

From Global Scores to Conditional Structural Activation

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Abstract

MAAT versions v1.2–v1.6 treated structural supports primarily as coordinates that can be compressed into scalar scores, defect-field functionals, or search-priority functions. Recent benchmarks in SAT search, two-qubit pointer robustness, and forced-fluid early warning reveal a recurring pattern: some supports do not behave like ordinary additive or multiplicative score terms. Instead, they decide when another structural mode should become active.

This note introduces MAAT v1.7 as a gate-aware refinement of the framework. The central change is the separation between *scoring coordinates* and *gating coordinates*. In particular, emergent robustness R_{rob} is no longer interpreted only as a contribution to a global score. It may act as a conditional activation variable:

$$F_{\text{active}}[X] = G_{\text{gate}}(R_{\text{rob}}[X]) F_{\text{score}}[H, B, S, V; X].$$

Equivalently, warning systems may use

$$W_{\text{gate}}[X] = \mathbf{1}[R_{\text{rob}}[X] < R_{\star}] f(S, V, A, P_{\text{tail}}; X).$$

A three-domain pilot re-analysis supports the gate direction but not a completed v2.0-style paradigm shift. SAT shows a small positive improvement from structure-gating (+2.77% mean compute-cost reduction relative to global Mode A). A scalar two-qubit pointer test shows a large in-distribution ranking gain (Spearman 0.2513 $\rightarrow$ 0.5150). A forced-2D fluid trigger shows improved utility (1.275 $\rightarrow$ 1.500), but only with threshold sensitivity and reduced event coverage.

MAAT v1.7 should therefore be read as a conservative framework update: structural supports should not always be compressed into one score. Some supports act as gates that determine when other structural modes become relevant.

1 Purpose

The purpose of this note is to formalize a pattern that appeared repeatedly in recent MAAT benchmarks. The pattern is not that MAAT scores always win. They do not. The pattern is subtler:

Global structural compression often loses information, while conditional activation can preserve when a structural mode is relevant.

This observation motivates MAAT v1.7:

Gate-Aware Structural Selection.

The update is intentionally modest. It does not replace v1.6 structural search geometry. It refines it by separating two roles that were previously often mixed:

- **score coordinates**, which rank candidates once a mode is active;
- **gate coordinates**, which decide whether that mode should act at all.

2 From Scores to Gates

Earlier MAAT score forms can be summarized schematically as

$$F_{\text{MAAT}}[X] = F(H, B, S, V, R_{\text{rob}}; X). \quad (1)$$

In support form, this often appears as a multiplicative or logarithmic compression:

$$F_{\text{MAAT}}[X] = - \sum_a \lambda_a \log(\epsilon + \Gamma_a[X]). \quad (2)$$

Such forms are useful when all supports contribute continuously to a single ranking problem.

However, several recent benchmarks indicate that this assumption is too strong. In some regimes, a support is not asking:

How much should this candidate be rewarded or penalized?

It is asking:

Should this structural channel be active here?

This is a different mathematical role. A score is graded. A gate is conditional.

3 v1.7 Definition

MAAT v1.7 introduces the following distinction. Let

$$X \in \Omega$$

be a candidate state, configuration, search node, trajectory, or diagnostic window. Let

$$H[X], B[X], S[X], V[X]$$

denote operational structural supports, and let

$$R_{\text{resp}}[X] = (H[X]B[X]V[X])^{1/3}, \quad R_{\text{rob}}[X] = \min\left(R_{\text{resp}}[X], (H[X]B[X]S[X]V[X])^{1/4}\right)$$

be the v1.2.1 robustness closure.

In v1.7, R_{rob} may enter as a gate:

$$\boxed{F_{\text{active}}[X] = G_{\text{gate}}(R_{\text{rob}}[X]; \theta) F_{\text{score}}[H, B, S, V; X].} \quad (3)$$

Here θ denotes a predeclared gate threshold or calibrated gate parameter.

Two simple choices are:

$$G_{\text{gate}}^{\text{hard}}(R_{\text{rob}}; R_{\star}) = \mathbf{1}[R_{\text{rob}} \geq R_{\star}], \quad (4)$$

$$G_{\text{gate}}^{\text{soft}}(R_{\text{rob}}; R_{\star}, \tau) = \frac{1}{1 + \exp[-(R_{\text{rob}} - R_{\star})/\tau]}. \quad (5)$$

For warning systems, the direction may reverse. Low robustness can activate a warning channel:

$$\boxed{W_{\text{gate}}[X] = \mathbf{1}[R_{\text{rob}}[X] < R_{\star}] f(S, V, A, P_{\text{tail}}; X).} \quad (6)$$

Thus high R_{rob} may gate in selection contexts, while low R_{rob} may gate in warning contexts.

4 Interpretation of Robustness

The v1.7 interpretation is:

Robustness is not always a score. Often it is a regime variable.

This resolves a recurring ambiguity. If R is treated as a fifth score-like primitive, it can double-count information already present in H, B, V . If R is removed entirely, the framework loses an important conditional signal. The gate interpretation avoids both problems:

R is real as an emergent diagnostic, R is not necessarily a primitive score coordinate.

In this sense, v1.7 extends the v1.2.1 hierarchy rather than reversing it. The primitive diagnostic supports remain H, B, S, V . Robustness remains emergent. Its new role is conditional activation.

5 Three-Domain Pilot

The v1.7 pilot reuses existing repository artifacts from:

- Paper 63: structure-gated Mode-A CDCL branching;
- Paper 64: two-qubit pointer-state robustness;
- Paper 67: forced-2D structural early warning.

No new external dataset is introduced. The pilot asks whether an R -gate improves over an R -score in each domain.

Domain	Score form	Gate form	Metric	Result
SAT	global Mode A	structure-gated Mode A	mean compute cost	20.5568 $\rightarrow$ 19.9865
Quantum	$HBSV \cdot R_{\text{rob}}$	$HBSV \cdot \mathbf{1}[R_{\text{rob}} \geq q_{75}]$	Spearman	0.2513 $\rightarrow$ 0.5150
Fluid	W_{MAAT}	base warning $\cdot \mathbf{1}[R_{\text{rob}} \leq q_{75}]$	utility	1.275 $\rightarrow$ 1.500

Table 1: Three-domain gate-hypothesis pilot. The signs of the gates differ by task: robust states are gated in for selection, while low-robustness states are gated in for warning.

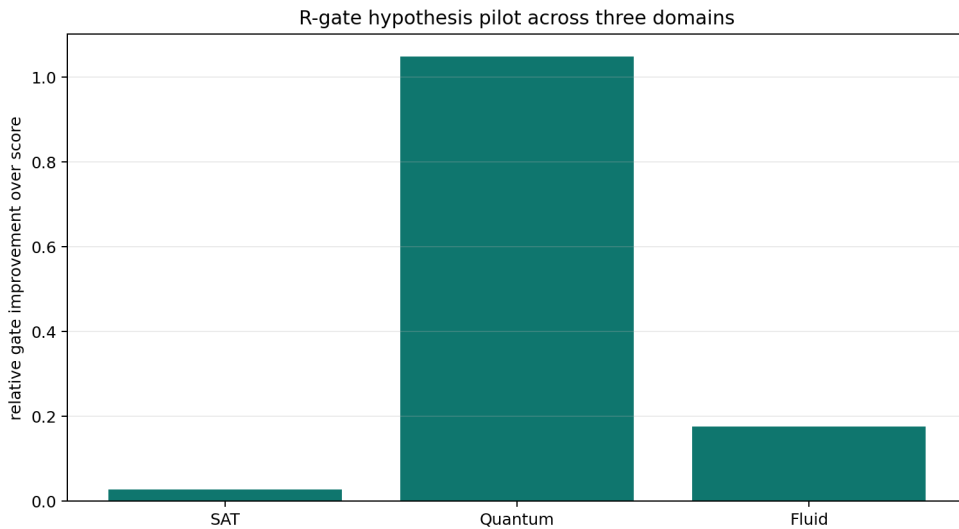


Figure 1: Relative improvement of gate form over score form in the three-domain pilot. The result supports v1.7 as a gate-aware refinement, not v2.0 as a completed paradigm.

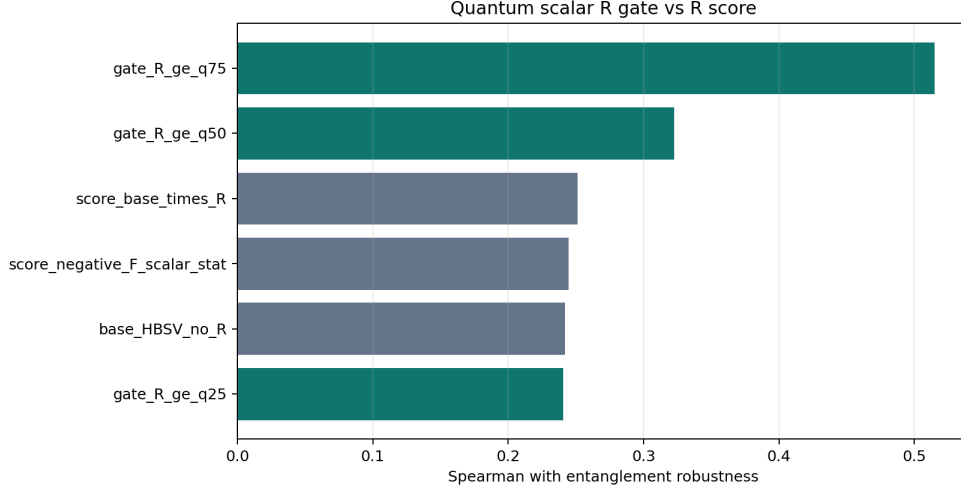


Figure 2: Quantum scalar gate test. A hard upper-robustness gate improves in-distribution ranking of two-qubit entanglement robustness over the scalar $HBSV \cdot R_{\text{rob}}$ score.

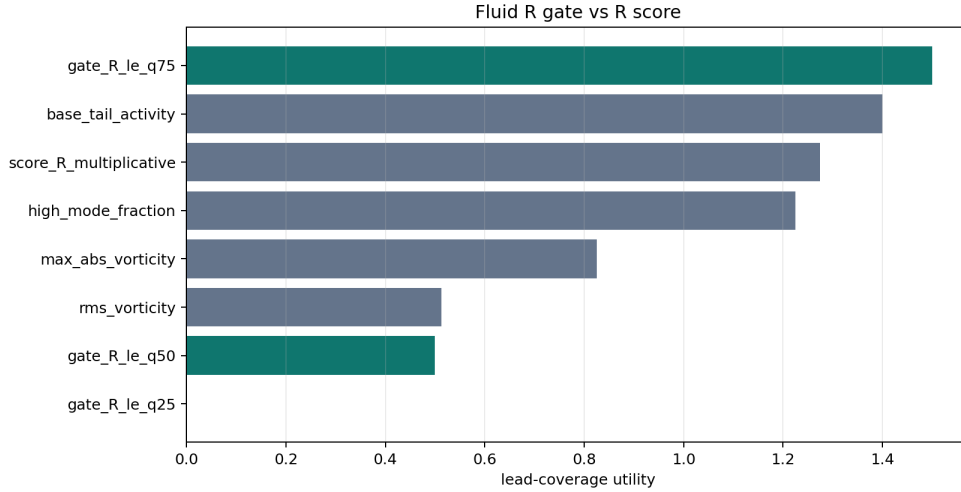


Figure 3: Fluid trigger gate test. The best hard low-robustness gate improves lead-coverage utility relative to the predeclared W_{MAAT} , but it sacrifices event coverage and is threshold-sensitive.

6 Why This Is v1.7, Not v2.0

The pilot is encouraging, but it is not a final theory. Several reasons prevent a v2.0 claim:

- The SAT effect is small and only weakly positive.
- The quantum gate is in-distribution and scalar, not a full field-model replacement.
- The fluid gate improves utility only under a selected threshold and with reduced event coverage.
- No external gate benchmark has yet been performed.
- Gate thresholds are not yet derived from a general calibration rule.

Thus v1.7 is the correct level of claim. It adds an important structural distinction:

score coordinate $\neq$ gate coordinate.

It does not yet provide a universal gate theory.

7 Practical Rule

A practical v1.7 workflow is:

1. define the candidate structural score without R ;
2. decide whether robustness should gate high- R states, low- R warning windows, or neither;
3. predeclare $R_\star$ or a calibration procedure;
4. compare score-only, score-with- R , and gate-with- R ;
5. report failures and threshold sensitivity explicitly.

The key methodological rule is:

A gate must be tested against a score. Otherwise “robustness” becomes an unfalsifiable explanatory word.

8 Limitations

- This note re-analyses internal artifacts only.
- The gate thresholds are simple quantile thresholds, not derived from a general theory.
- The three domains use different metrics: compute cost, correlation, and trigger utility.
- The pilot does not prove that R is always a gate.
- The pilot does not justify abandoning scalar MAAT scores; it only shows that conditional activation should be treated as a first-class option.

9 Reproducibility

Repository:

<https://github.com/Chris4081/structural-selection-principle>

Run:

```
cd experiments/maat_v17_gate_hypothesis_pilot
python3 gate_vs_score_pilot.py
```

Generated outputs:

- v17_gate_hypothesis_summary.csv,
- v17_gate_hypothesis_summary.json,
- sat_gate_instance_deltas.csv,
- sat_gate_family_deltas.csv,
- quantum_gate_scalar_results.csv,
- fluid_gate_trigger_results.csv,
- three PNG figures.

Data and attribution note. This paper introduces no new external dataset. It reuses derived internal artifacts from Papers 63, 64, and 67. Those underlying papers contain their own dataset and no-endorsement notes.

10 Conclusion

MAAT v1.7 introduces gate-aware structural selection. Its central claim is narrow:

Structural supports should not always be compressed into one score; some supports act as gates that decide when other structural modes become relevant.

The current three-domain pilot supports this update. SAT gives a weak positive signal. Quantum gives a strong in-distribution scalar signal. Fluid gives a positive but threshold-sensitive trigger signal.

This justifies MAAT v1.7 as a framework refinement. It does not yet justify v2.0. The next step is a predeclared gate-validation paper with calibrated thresholds, external benchmarks, and explicit failure criteria.

References

- [1] C. Krieg, “Structure-Gated Mode-A CDCL Branching,” preprint (2026).
- [2] C. Krieg, “Two-Qubit Pointer Selection,” preprint (2026).
- [3] C. Krieg, “Structural Early Warning as a Refinement Trigger,” preprint (2026).